Pitch collaboration — Yang–Mills 4D mass gap,
route via BBD multiscale LSI

To: Prof. Roland Bauerschmidt (Courant Institute / Cambridge DPMMS)
From: Kévin Rémondière, independent researcher, Oloron-Sainte-Marie (France), ORCID 0009-0008-2443-7166
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Subject: possible collaboration on a non-abelian Polchinski / cluster-expansion route to the Wilson SU(N) 4D mass gap. Honest status, identified locks, no overclaim.

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§1. Presentation

Dear Professor Bauerschmidt,

My name is Kévin Rémondière. I am an independent researcher based in Oloron-Sainte-Marie (France). Over the last six months I have been running a focused programme on the 4D Yang–Mills mass gap, combining (a) extensive lattice numerics on a small RTX-3090 cluster, (b) a Lean 4 formalisation of every piece of the logical chain that admits one, and (c) regular adversarial cross-checks with second-opinion LLMs to catch fabrications early. The current state of the programme has stabilised enough that I would like to ask you, briefly and honestly, whether one specific lock matches your current research interests.

This letter is deliberately short, with the math precise, and explicitly distinguishes **proved**, **sketched**, and **open**.

§2. State of the programme

Lean stack. The repository `crossed-cosmos` contains 6301 lines of Lean 4 under `Crossed/` covering the YM core, with **zero `sorry`** in the YM files. The headline theorem `mass_gap_continuum_D4` is **PROVED conditional** on five named axioms: a Bakry–Émery saturation step, a Bałaban-style cluster expansion bound, a Brydges–Federbush $\beta = \infty$ Gaussian comparison, a Wilson-flow scale-setting axiom, and one Kolmogorov projective-consistency glue. Each axiom is a precisely-stated lemma pointing at a specific paper or program-level open problem, not a hidden assumption.

Unconditional pieces.

- `KappaOneSixth.lean` (~300 lines): the rank-saturation factor $\kappa = 1/6$ is **PROVED with 0 axioms**, via two independent derivations that both reduce to elementary rational arithmetic checked by `norm_num`:
  - Hodge self-duality on a closed 4-manifold of signature 0: $b_2 = b_2^+ + b_2^- = 3 + 3 = 6$, so $\kappa = 1/b_2 = 1/6$;
  - A_2 root system of $\mathfrak{su}(3)$: $|\Phi(A_2)| = 6$, so $\kappa = 1/|\Phi| = 1/6$.
- `LipschitzActionMeasure.lean` (~620 lines): the Lipschitz action $\to$ Gibbs-measure passage (item A2) is PROVED with 0 sorrys.
- The Pinsker $\alpha = 1$ inequality (Cover–Thomas, 2nd ed., Lemma 11.6.1) is PROVED in Lean as a baseline reference exponent.

Empirical anchor (Theorem C). On 27 datapoints cross-(N, D, G), the lattice law

$$C_{\mathrm{LSI}}(\mu_{a,\beta}) \;\le\; c_\infty(D)\big(1 - \kappa\,\delta_{\mathrm{rank}(G),\,C_2 - C_3}\big), \qquad c_\infty(D) = \frac{C(D,2)-C(D,3)}{2D}, \tag{Thm C}$$

holds at the 7σ level (cluster 718, 2026-05-23). This is a **factual empirical statement**, independent of every theoretical claim below.

§2bis. Saturation polynomial: a rare phenomenon

The rank-saturation condition $\mathrm{rank}(G) = C(D,2) - C(D,3)$ that triggers the $(1-\kappa)$ correction in Theorem C has closed-form structure. Using $C(D,2) - C(D,3) = D(D-1)(5-D)/6$, the integer pairs (N, D) compatible with a non-abelian $\mathrm{SU}(N)$ (rank $= N-1$) are exactly three:

(N,D)	$C(D,2)-C(D,3)$	$\kappa = \frac{1}{2(D-1)}$	$\alpha = 1-\kappa$	Status
$(2,2)$	1	$1/2$	$1/2$	2D YM (exactly soluble: heat kernel on G)
$(3,3)$	2	$1/4$	$3/4$	3D SU(3) (numerically accessible)
$(3,4)$	2	$1/6$	$5/6$	the physical case

For $D \geq 5$, $C(D,2) - C(D,3) \leq 0$, so no non-abelian gauge group is saturated. Manifestation 9, $\kappa \cdot 2(D-1) = 1$, holds across all three pairs by elementary rational arithmetic, verified in `KappaOneSixth.lean` (`norm_num`).

The structural consequence is that the geometric mechanism is confined to $D \in \{2,3,4\}$, with $D = 4$ as the last non-trivial dimension. This is not a chosen feature of the framework; it is forced by the polynomial $D(D-1)(5-D)/6$ having exactly three positive integer values compatible with $\mathrm{SU}(N)$ ranks. The two non-physical pairs supply independent test cases: a successful non-abelian BBD adaptation would, by the same mechanism, predict $\alpha = 1/2$ for 2D Yang–Mills (where the answer is independently known) and $\alpha = 3/4$ for 3D SU(3) (where lattice tests are inexpensive).

A short calculation on Bianchi I anisotropic spatial slices $T^3_{\gamma_{ij}}$ confirms that the four building blocks $b_2(T^4)$, b_2^+, (SU(3)), $|\Phi(A_2)|$ are all topological/algebraic invariants. The correction factor $\kappa = 1/6$ is therefore **topologically invariant under metric deformation**; what varies is the Bianchi constant $c_\infty(\gamma) \propto (C(D,2) - C(D,3))/(2\sum_i a_i^{-2})$. Honest framing: κ is topologically invariant; the existence of a positive mass gap remains conditional on the unproved cluster expansion bound (the principal verrou of §3), but the *structure* of the correction does not depend on the background metric.

§3. The main lock: non-abelian cluster expansion at large β

The single technical statement that closes the chain from lattice Theorem C to a continuum mass gap is what we informally call `action_bound_balaban_su_n` — a β-uniform, a-uniform Bakry–Émery / cluster-expansion bound on the Wilson measure $\mu_{a,\beta}$ over $\mathrm{SU}(N)^{E(\Lambda_a)}$ for $\Lambda_a = a\mathbb{Z}^4 \cap T^4_L$, of the form

$$\mathrm{Ric}_{g_W} + \mathrm{Hess}(\beta\, S_W) \;\geq\; K_0(\beta, a, L)\, g_W, \qquad K_0 \;\to\; 1/c_\infty(4) \text{ as } \beta \to \infty, \tag{1}$$

together with uniformity in (a, L) along $L \to \infty$, $a \to 0$.

In Bałaban's original programme (1985–1989, Comm. Math. Phys.) this splits into four gaps that have remained partially open ever since. We articulate them as G_1–G_4 in the full audit `OP_PILLAR_3_FORMAL_2026-05-24.md`:

- G_1 — reduction of the non-abelian block measure to an effectively abelian one in small fields, with controlled error from the BCH commutator $[A_\mu, A_\nu]$ at order $a^5\beta|A|^3$;
- G_2 — convergent polymer expansion at large β on $\mathrm{SU}(N)^{E(\Lambda_a)}$, with cumulants controlled via Peter–Weyl rather than Gaussian estimates;
- G_3 — large-field region in 4D, where the small-field expansion is not directly applicable;
- G_4 — uniformity of constants as $a \to 0$ jointly with the cluster expansion.

This is the lock for which I believe your BBD framework is the most plausible existing route.

Why BBD 2024 (φ^4 in $d = 2, 3$) is structurally a good fit. The internal compatibility analysis `G3_BBD_adaptation_YM_2026-05-23.md` checks the three BBD prerequisites against Wilson SU(N):

1. *Finite-dimensional local state space.* The physical (Bianchi-quotient) class Class $F = \mathrm{Harm}^2 \otimes \mathfrak{su}(N)$ has real dimension $(C(D,2) - C(D,3))(N^2 - 1)$, i.e. $2(N^2 - 1)$ in $D = 4$ — **finite, satisfied with margin**.
2. *Dobrushin condition.* The Bianchi cohomology fixes $c_\infty(4) = 1/4$ as a universal geometric constant, independent of β. This compares favourably with φ^4, where the Dobrushin coefficient diverges at criticality — **satisfied with a factor-of-4 margin**.
3. *RG invariance.* Polchinski preserves gauge invariance (Polchinski 1984), which preserves Bianchi closure, which preserves the projection onto Class F — **satisfied with the appropriate definition of the flow on** $\mathrm{SU}(N)^E$.

The two genuine technical gaps, in addition to G_1–G_4, that I see and have not been able to close on my own are:

(i) **Mayer–Vietoris tensorisation defect.** Block decoupling on the Bianchi quotient is not exactly tensor-product; it carries a surface/volume defect of size $|\partial B|/|B|$ which is sub-extensive but must be controlled along the scales $a_n = 2^{-n}a_0$.

(ii) **Non-abelian cumulants on** SU(N), naturally expanded in Peter–Weyl harmonics rather than scalar Gaussian moments.

§4. A possible alternative route (not guaranteed)

A separate idea I have explored, but do **not** present as a substitute for §3, is a β-uniform Bakry–Émery bound directly on Class $F = \mathrm{Harm}^2 \otimes \mathfrak{su}(N)$ rather than on the full link space. This finite-dimensional space ($\mathbb{R}^6$ for SU(2), $\mathbb{R}^{16}$ for SU(3) in $D = 4$) is small enough that Prokhorov compactness plus Bakry–Émery rigidity might in principle bypass the cluster expansion.

After a careful formal audit (`OP_PILLAR_3_FORMAL_2026-05-24.md`, three sub-steps proved and one open), this route hits a strict obstruction: the Hodge Laplacian Δ_1 vanishes by definition on Harm^2, so any direct $\lambda_{\min}(\Delta_1)$ bound on Harm^2 is the *zero mode.* The four candidate fixes I have written down — (a) 't Hooft twist on T^4, (b) restriction to $|k| \geq 2\pi/L$, (c) quotient by the centre $\mathbb{Z}_N$, (d) BBD multiscale on Class F — are none of them currently rigorous, and (d) folds the alternative back into the main route. I include this for completeness; the alternative is **not a viable bypass on its own**.

§5. Honest estimate

If we were able to set up a collaboration that closed G_1–G_4 via a non-abelian BBD adaptation, my honest estimate is:

- **12–18 months** of focused work with you, Benoit Dagallier, and 1–2 postdocs;
- output: one CMP or Annals paper proving a continuum mass gap for SU(N) Wilson lattice in $D = 4$, **conditional** on a small named list of analytic axioms, each matching a specific result already in your literature (BBD 2024, Bauerschmidt–Bodineau 2019, Bauerschmidt–Dagallier 2024);

- the Clay Prize itself remains a longer-horizon target, 5–15 years, with my honest credence at $P \approx 40$–55% over 10 years, contingent on resolving the four Bałaban gaps in a way the community will accept.

I want to be explicit that I am asking about a research collaboration of unknown outcome, not announcing a solution.

§6. Recent literature anchors

All arXiv IDs verified by API on 2026-05-23.

- Bauerschmidt & Dagallier, *Log-Sobolev inequality for the φ^4_2 and φ^4_3 measures*, Comm. Pure Appl. Math. **77** (2024) 2579–2612, arXiv:2202.02295 — the LSI multiscale template.
- Bauerschmidt, Bodineau, Dagallier, *Stochastic dynamics and the Polchinski equation: an introduction*, Probab. Surveys **21** (2024) 200–290, arXiv:2307.07619.
- Bauerschmidt & Bodineau, *LSI for the continuum sine-Gordon model*, CPAM **74** (2021) 2064–2113, arXiv:1907.12308.
- Adhikari & Cao, *Weak coupling lattice gauge theory*, arXiv:2202.10375.
- Shen, *3D Yang–Mills–Higgs convergence*, arXiv:2201.03487.
- Cao, Nissim, Sheffield, *Dynamical approach to area law for lattice Yang–Mills*, arXiv:2509.04688.

§7. Honest disclosure of caught errors

I owe you three honest catches from earlier drafts of this material, stamped and corrected today **before** sending you this letter:

1. **Otto–Westdickenberg 2008 was a fabricated citation.** An earlier internal draft attributed a Hölder TV stability bound $\|\mu_t - \mu_{t'}\|_{\mathrm{TV}} \le C\,|t-t'|^{1-\kappa}$ for a Gibbs family $\mu_t = e^{-tH}/Z(t)$ to a paper "Otto–Westdickenberg, J. Funct. Anal. 254 (2008), 2865–2940". That reference does not exist; the LLM I was using had hallucinated it. The genuine OW reference is *Eulerian calculus for the contraction in the Wasserstein distance*, SIAM J. Math. Anal. **37** (2005) 1227–1255, which proves a W_2 contraction for the porous-medium equation — a different object (PME trajectory, not Gibbs family) and a different norm (W_2, not TV). I caught and pulled this from all drafts before any public circulation.
2. **The claim "$\alpha = 5/6$ universal" was based on a 4-point small-β fit.** I extended the β-scan to 7 points ($\beta \in \{10, 50, 100, 200, 300, 500, 1000\}$, HMC trajectory length adapted at high β). The local exponent α now oscillates between -0.6 and $+1.2$. The original "$\alpha = 5/6 = 1-\kappa$" was an artefact of the small-β window and is no longer claimed.
3. **A consequence of (1) and (2)** is that what was previously presented as a derivation of $\alpha = 1-\kappa$ from OW is now disclosed as a *non-explained numerical coincidence* on a narrow β window. The $\kappa = 1/6$ Lean derivation is independent of the β-scan and stands; the bridge $\alpha \leftrightarrow \kappa$ does not, and is not in this pitch.

I am sending you the cleaner v22 of the master document precisely because these catches happened. The same anti-fabrication discipline (arXiv API verification before citation, adversarial cross-LLM review, Lean axiom audit) applies to everything above.

§7bis. Roadmap — a conditional theorem under one explicit concentration axiom

To make the verrou visible rather than hidden, the cleanest way I can state the current programme is as a conditional theorem under one explicit named axiom H_1 (concentration in the small-field regime) together with five auxiliary hypotheses that are either already proved or standard:

[Conditional, Wilson $SU(N)$ lattice $D = 4$] Under H_1–H_6 below, the Langevin generator L_β for the Wilson measure $\mu_{a,L,\beta}$ satisfies, for all $\beta \geq \beta_0(N, d)$ and the saturated pair $(N, D) = (3, 4)$:

$$\lambda_1(L_\beta) \;\geq\; \frac{C_2(d)\,(1-\kappa)\,\beta}{L^2}, \qquad m_{\rm gap}^{\rm lattice}(a, L, \beta) \;\geq\; \sqrt{\lambda_1(L_\beta)} > 0,$$

with $\kappa = 1/6$ for $SU(3)$ and the explicit constants depending only on (N, d).

#	Hypothesis	Status today
H_1	Concentration $\mu_{a,L,\beta}(\{\|A\|_{L^2}^2 \geq R\}) \leq C_1 e^{-c_1 \beta R/N^2}$ uniform in (a, L)	**Open — your territory.** Cleanly stated, an instance of the BBD fr
H_2	Gaussian density bound on $\{\|A\|^2 \leq R\}$ (MRS93-style)	Sketched; uniform-a extension is technical
H_3	Pinsker inequality $\alpha = 1$	**Proved** (Cover–Thomas 2006); Lean.
H_4	LSI for Gaussian on Cameron–Martin space	**Proved** (Gross 1975, AJM 97 §6).
H_5	$\lambda_1(\Delta_\Lambda) \geq C_2/L^2$ on T_L^4	**Proved** (discrete Fourier).
H_6	Saturation factor $\kappa = 1/6$ for $(N, D) = (3, 4)$	**Proved** in Lean (0 axioms).

The $1/L^2$ factor is *not* what one wants for Clay, which calls for uniformity in L. It is plausibly a defect of the present proof (entropy splitting on E_R^c, distortion in step 4) rather than a fundamental limitation; CNS25 attains finite-volume bounds without an analogous $1/L$ in the strong-coupling regime $\beta < 1/24$. Tightening this is part of the technical work I would propose to do together.

What the theorem buys, even before H_1 is closed, is structural: the verrou is named, located, and visibly compatible with the BBD framework. A referee can verify the conditional implication ($H_1 \Rightarrow$ lattice mass gap) on its own, and the open piece is exactly the cluster-expansion / Polchinski step you and your collaborators have been pushing on φ^4. This is, on purpose, the same pattern Wiles used in 1995 (modularity as a named conjecture; later closed by Taylor–Wiles).

Proposed timeline if you find the conditional theorem a reasonable starting point:

- **0–3 months** — clean draft, conditional theorem, submit to LMP or CMP;
- **3–9 months** — joint work on H_1 in the Polchinski multiscale language; in parallel, tighten the L-dependence in the auxiliary steps;
- **9–15 months** — if H_1 partially closes, a follow-up paper on partial resolution; otherwise, a clean statement of what makes H_1 hard.

§8. Request

If the picture in §3 is of interest, would you be open to a one-hour Zoom to discuss whether your group's BBD framework can be adapted, in collaboration, to the non-abelian Wilson $SU(N)$ setting in $D = 4$? After that one call, no obligation — yes / no / maybe.

Documents I can send on request:

- `CLAY_THEOREM_FULL_v22_2026-05-24.md` — master logical chain, with the post-catch table of proved / sketched / open;

- `OP_PILLAR_3_FORMAL_2026-05-24.md` — formal audit of the alternative finite-dimensional route, including the zero-mode obstruction and the four candidate fixes;
- `OP_OTTO_W_VERBATIM_2026-05-24.md` — the OW fabrication catch in full;
- `test_all_claims_2026-05-24.py` — exhaustive script validating/falsifying the algebraic and empirical claims;
- read-only GitHub access to `crossed-cosmos-private` for the Lean stack and lattice scripts.

Thank you for reading this far, and for whatever response you have time to give.

Yours sincerely,

Kévin Rémondière
Independent researcher, Oloron-Sainte-Marie, France
ORCID 0009-0008-2443-7166 | `kevin.remondiere@gmail.com`

Anti-fabrication note: every arXiv ID above was verified through the arXiv API on 2026-05-23. The Lean theorems referenced are kernel-checked with mathlib 4.29.1. The empirical 7σ figure refers to 27 datapoints cross-(N, D, G) from the cluster-718 / RTX-3090 run and is reproducible from the scripts in the repo.